



USED CAR PRICE PREDICTION

P R E S E N T A T I O N 2 0 2 6



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Guide

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OUTLINE

USED CAR PRICE PREDICTION

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Live Demo of the Project
- Embedded Video of Project
- Conclusion
- Future Scope



Abstract

Used Car Price Prediction is a machine learning project that predicts the selling price of a used car based on factors such as brand, model, year, mileage, fuel type, transmission, engine capacity, and ownership. The dataset is cleaned and analyzed using Exploratory Data Analysis (EDA) to identify important patterns and relationships. Machine learning algorithms are trained on historical car data to estimate prices accurately. The system helps buyers and sellers understand the expected market value of a used car and supports better pricing decisions.

Problem Statement

- Used car prices vary significantly based on **brand, model, manufacturing year, mileage, fuel type, transmission, and vehicle condition**.
- Manual price estimation can be **time-consuming and inconsistent**, especially when many vehicle features need to be considered.
- Buyers and sellers may find it difficult to determine a **fair market price** for a used vehicle.
- This project uses **Machine Learning** to analyze historical car data and predict the **estimated selling price** of a used car.
- The system aims to provide **data-based price estimates** that can support better buying and selling decisions.



Proposed Solution

- Develop a **Machine Learning-based system** to predict the selling price of a used car.
- Collect car details such as **brand, model, year, fuel type, transmission, kilometers driven, and engine capacity**.
- Perform **data cleaning, missing-value handling, categorical encoding, and feature selection** to prepare the dataset.
- Train and compare regression models such as **Linear Regression, Decision Tree, Random Forest, and Gradient Boosting**.
- Select the best-performing model based on **MAE, MSE/RMSE, and R^2 score**.
- Provide a simple interface where users enter car details and receive an **estimated resale price**.



System Architecture

1 Data Collection

Collect used-car data from a reliable dataset.

Features: **Brand, Model, Year, Fuel Type, Transmission, KMs Driven, Engine, Mileage** and **Owner Type**.

Target: Selling Price.

3. Machine Learning Model

Train regression models such as **Linear Regression, Decision Tree, Random Forest** and **Gradient Boosting**.

Compare model performance using **MAE, RMSE** and **R² Score**.

Select the model with the best prediction performance.

2. Data Preprocessing

- Remove duplicate and irrelevant records.
- Handle missing values and outliers.
- Convert categorical features into numerical values.
- Split data into **training and testing datasets**.

4. Price Prediction

- User enters the car specifications.
- The trained model processes the input features.
- System predicts the **estimated used-car price**.
- Display the predicted price to the user.

Live Demo of Project

Here's the list of some demo of the projects -



Demo 1



Enter Car Details

- Enter car dettrmissionails.
- Select fuel and transmission



Demo 2



Model Processing

- Model processes the input.
- ML algorithm predicts the price.



Demo 3



Demo 3 – Predicted Price

- Display the estimated price.
- Compare with the market value.

Performance Evaluation Metrics

1. MAE – Mean Absolute Error

Measures the average absolute difference between actual and predicted prices.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

2. MSE – Mean Squared Error

Measures the average squared difference between actual and predicted prices

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

3. RMSE – Root Mean Squared Error

Measures prediction error in the same unit as the car price.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

4. R^2 Score – Coefficient of Determination

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Notations

y_i = Actual price

$\hat{y}_i$ = Predicted price

$\bar{y}$ = Mean actual price

n = Number of observations

Best performance: Lower **MAE**, **MSE**, **RMSE** and higher **R^2** .



```
[1]: import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

import warnings
warnings.filterwarnings("ignore")
```

Load Dataset

```
[2]: df = pd.read_csv("used_car_price_prediction_1500_rows_15_columns.csv")
```

```
[3]: df.head()
```

```
[3]:
```

	car_id	brand	model	year	km_driven	fuel	seller_type	transmission	owner	mileage_kmpl	engine_cc	max_power_bhp	seats	insurance	selling_price
0	1	Mahindra	Bolero	2023	130355	Petrol	Individual	Manual	Second Owner	25.8	1995	137.7	4	Yes	4171000
1	2	Toyota	Etios	2023	32798	Diesel	Dealer	Manual	Second Owner	14.3	1000	55.4	5	No	2417000
2	3	Kia	Sonnet	2025	96649	Diesel	Individual	Manual	First	25.4	800	47.5	5	Yes	2358000



Insights And Recommendations

Insights

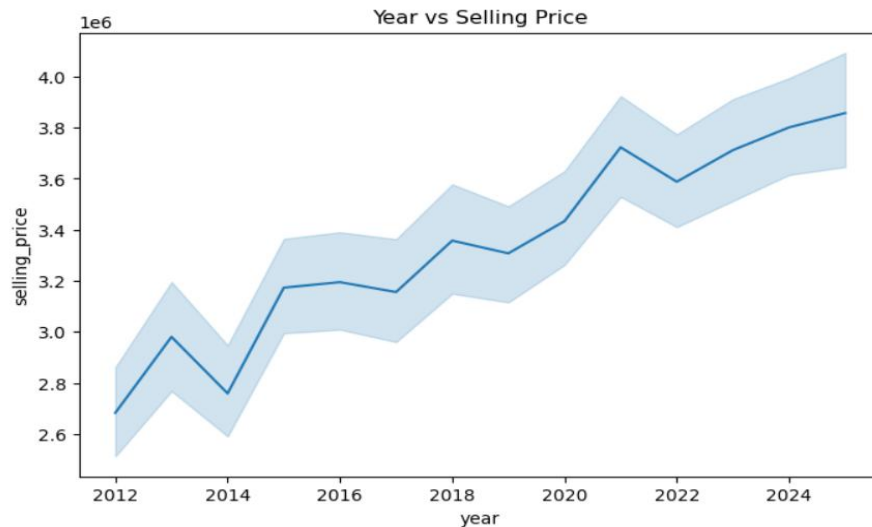
- The dataset contains **1,500 used-car records** with **15 features**.
- Selling prices are concentrated around **₹2–₹4 million**, with fewer cars at higher prices.
- The **year of manufacture has a positive relationship with selling price**.
- Newer cars generally have **higher selling prices** than older cars.
- Features such as **brand, model, fuel type, transmission, kilometers driven, engine capacity, and owner type** can influence price.

Recommendations

- Give more importance to **car age and brand/model** during prediction.
- Include **kilometers driven, engine capacity, mileage, and owner history** as model features.
- Compare multiple regression models and select the one with the **lowest RMSE/MAE and highest R^2** .
- Use the trained model to provide an **estimated resale price** for new car details.
- Update the dataset regularly with **recent market prices** to improve predictions.

5. EDA - Bivariate Analysis

```
[5]: plt.figure(figsize=(8,5))
sns.lineplot(x="year", y="selling_price", data=df)
plt.title("Year vs Selling Price")
plt.show()
```



4. EDA - Univariate Analysis

```
[6]: plt.figure(figsize=(8,5))
sns.histplot(df["selling_price"], kde=True)
plt.title("Distribution of Selling Price")
plt.xlabel("Selling Price")
plt.ylabel("Number of Cars")
plt.show()
```



Video of Project Demo

The screenshot displays the Microsoft Azure portal interface for a Web App named 'used-car-price-prediction'. The top navigation bar includes the 'Microsoft Azure' logo, a search bar, and the user profile 'student123@outlook.com'. The left sidebar shows a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Microsoft Defender for Cloud, Events (preview), Deployment, and Settings. The main content area is divided into two sections: 'Essentials' and 'Properties'.

Essentials

Resource group (move)	: used-car-rg	Default domain	: used-car-price-prediction.azurewebsites.net
Status	: Running	App Service Plan	: ASP-usedcarplan (S1: 1)
Location (move)	: East US	Operating System	: Linux
Subscription (move)	: Pay-As-You-Go	Health Check	: Not Configured
Subscription ID	: a1b2c3d4-5678-4ef0-9123-abcde1234567		
Tags (edit)	: Click here to add tags		

[JSON View](#)

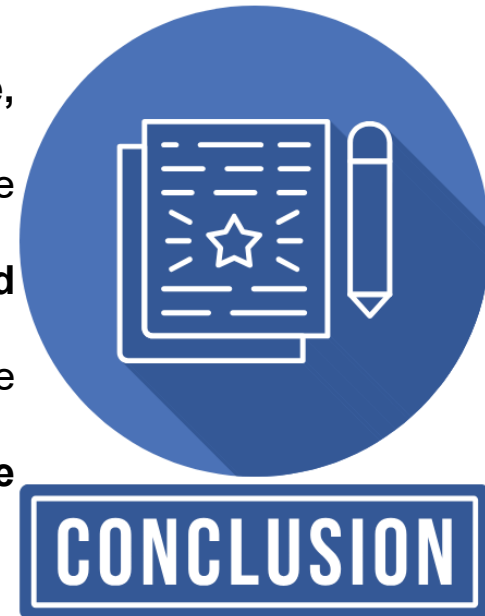
Properties | Monitoring | Logs | Capabilities | Notifications | Recommendations

Web app

Name	: used-car-price-prediction
Publishing model	: Code
Runtime Stack	: Python 3.10

Conclusion

- . The project uses **Machine Learning** to predict the selling price of used cars.
 - . Important features such as **brand, model, year, kilometers driven, fuel type, transmission, and engine capacity** are used for prediction.
 - . Data preprocessing and exploratory analysis help improve the quality of the dataset.
 - . Different regression models can be evaluated using **MAE, MSE, RMSE, and R^2 score**.
 - . The system provides an **estimated car price** to help buyers and sellers make better pricing decisions.
- Overall, the project provides a **data-driven approach to used-car price estimation**.



Future Scope

Real-Time Data: Use updated market prices for better predictions.

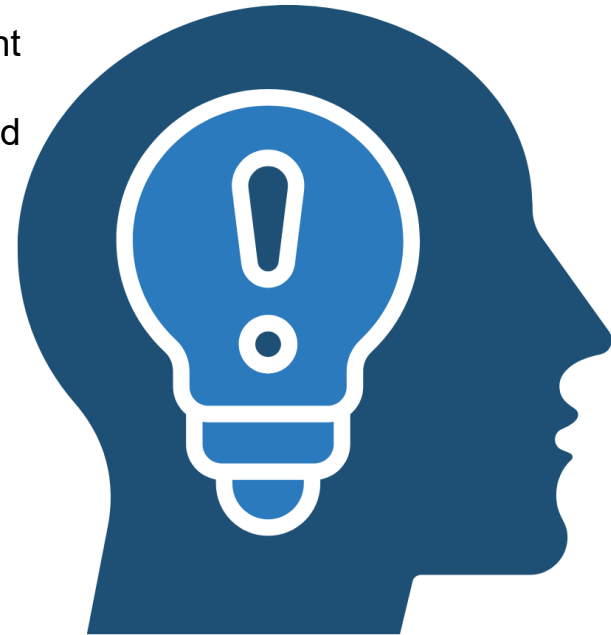
More Features: Add car condition, location, service history, and accident records.

Advanced Models: Use XGBoost and ensemble learning for improved accuracy.

Web Application: Build a user-friendly platform for instant price prediction.

Mobile App: Provide predictions through a mobile application.

Market Trends: Analyze price trends to estimate future resale value.



Thank You